Scenario Identification Module (SCIM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Governed Space

Simulation Scenario Identification

Canonical Definition

Scenario Identification Module (SCIM) defines the governance framework for identifying, distinguishing, classifying, bounding, documenting, and maintaining the materially relevant scenarios that a simulation may need to examine in order to satisfy its defined Simulation Purpose & Scope.

SCIM establishes the first internal governance space of the Simulation Scenario Standard (SSS).

Its role is not to generate every mathematically possible situation.

Its role is to establish a governed answer to a more important question:

Which conditions are sufficiently relevant to become identifiable Simulation Scenarios rather than remaining hidden, assumed, ignored, or informally understood?

Operational Role

SCIM receives the governed simulation basis established by preceding SIMULOS® Parent Standards:

Simulation Object

↓

Simulation Purpose & Scope

↓

Reality Representation

↓

Simulation Assumptions

↓

Simulation Model & Environment

↓

Simulation Inputs & Data

SCIM then establishes:

Scenario Identification Basis → Candidate Conditions → Materiality Assessment → Scenario Recognition → Scenario Identity → Scenario Classification → Scenario Register

The resulting identified scenarios become the governed basis for subsequent:

Scenario Variation → Edge Cases → Stress Conditions → Scenario Coverage

SCIM therefore governs what enters the Scenario Design space.

Module Operational Space

SCIM governs: scenario requirements, candidate scenario identification, scenario recognition, scenario relevance, scenario materiality, scenario identity, scenario boundaries, scenario distinction, scenario classification, scenario origin, scenario rationale, scenario dependencies, scenario relationships, scenario applicability, scenario inclusion, scenario exclusion, scenario duplication, scenario families, scenario discovery, emergent scenarios, scenario lifecycle status, scenario version continuity, and Scenario Register formation.

Scenario Identification Basis

Scenario identification should begin from the governed simulation inquiry rather than from an arbitrary list of situations.

Potential scenario requirements may originate from: Simulation Purpose, Simulation Scope, represented reality, system operating states, Simulation Assumptions, Input Sets, parameter ranges, environmental conditions, dependencies, uncertainty, boundaries, known failure mechanisms, operational transitions, interactions, historical events, observed incidents, or simulation-specific exploratory questions.

The objective is to establish why a scenario belongs in the simulation inquiry.

Scenario Requirement

A Scenario Requirement identifies a condition, state, event, interaction, variation, boundary, failure, uncertainty, or other circumstance that may need deliberate simulation examination.

It answers:

What materially relevant condition must the Scenario Design process consider?

A Scenario Requirement does not yet require a fully configured executable scenario.

It creates the governed requirement from which one or more Simulation Scenarios may later be designed.

Requirement-to-Scenario Separation

SCIM distinguishes:

What must be examined

from:

How it will be simulated.

For example:

Requirement: Examine operation under materially degraded communication.

The eventual Scenario Set may contain several distinct communication-degradation scenarios.

This separation prevents one convenient technical scenario from being mistaken for complete satisfaction of the underlying scenario requirement.

Candidate Scenario

A Candidate Scenario is a potential simulation condition identified for governance assessment before formal inclusion in the Scenario Register.

Candidate scenarios may arise from: architecture analysis, domain knowledge, operational experience, historical incidents, system design, risk analysis, uncertainty analysis, simulation results, expert input, or automated scenario discovery.

Candidate status means:

potentially relevant — not yet governed as required.

Scenario Materiality

Not every conceivable condition deserves scenario status.

SCIM evaluates whether a candidate condition is materially relevant to the Simulation Purpose & Scope.

Materiality may depend upon whether the condition could materially affect: system behavior, simulation outputs, system-state transitions, performance, safety, failure behavior, decision behavior, resource consumption, interaction behavior, recovery, or downstream interpretation.

Materiality Is Not Frequency

A condition may be rare and still be material.

A condition may be common and still be immaterial to the particular simulation inquiry.

SCIM therefore separates:

How often might this occur?

from:

Could this condition materially change what the simulation needs to understand?

Scenario Recognition

A candidate condition becomes a governed Simulation Scenario when it has sufficient: relevance, identity, boundary, rationale, and connection to the Simulation Purpose & Scope

to require deliberate treatment within Scenario Design.

Recognition does not mean the scenario has already been executed.

It means the scenario has become a governed object that cannot silently disappear from the simulation inquiry.

Scenario Identity

Every material Simulation Scenario should be distinguishable from materially different scenarios.

Scenario Identity may include: Scenario Identifier, scenario name, Scenario Requirement, scenario purpose, applicable Simulation Object, applicable object version, relevant Simulation Scope, scenario condition, scenario state, relevant environment, relevant inputs, relevant parameters, dependencies, applicable assumptions, scenario origin, scenario version, and current scenario status.

Scenario Identity Principle

SIMULOS® establishes:

A scenario that cannot be uniquely identified cannot be reliably executed, compared, reproduced, changed, or traced.

Scenario Boundary

Every identified scenario should establish what belongs to that scenario and what does not.

A Scenario Boundary may distinguish: included conditions, excluded conditions, active system states, applicable environment, applicable actors, applicable dependencies, temporal boundaries, event boundaries, and configuration boundaries.

This prevents scenario meaning from changing informally between design and execution.

Scenario Distinction

Two scenarios should remain separate where a difference between them could materially affect simulation behavior or interpretation.

For example:

Communication Lost Before Autonomous Handover

and:

Communication Lost After Autonomous Handover

may require separate scenario identities where sequence materially changes system behavior.

Scenario Equivalence

Two candidate scenarios may be treated as equivalent where their differences are not material to the Simulation Purpose.

Equivalence should be justified rather than assumed.

Conceptually:

Candidate A + Candidate B → Material Difference Assessment → Separate Scenarios / Scenario Equivalence Scenario Classification

Identified scenarios may be classified to support systematic Scenario Design.

Classification may include:

Baseline

Nominal

Expected

Alternative

Variation

Boundary

Edge

Rare

Stress

Extreme

Failure

Degraded State

Transition

Recovery

Uncertainty-Driven

Combined

Cascading

Emergent

or another simulation-specific class.

Classification does not determine scenario importance by itself.

Multi-Class Scenarios

A scenario may belong to more than one class.

For example:

Rare + Extreme + Transition + Failure

may legitimately describe one scenario.

SCIM should preserve such relationships rather than forcing every scenario into one exclusive category.

Scenario Origin

A scenario should remain traceable to why it was identified.

Possible origins include:

Simulation Purpose

Scope Requirement

Reality Representation

Assumption

Input Uncertainty

System Boundary

Known Failure

Historical Incident

Operational Observation

Dependency

Previous Simulation Result

New Evidence

or:

Scenario Discovery

Scenario Origin improves later assessment of why the scenario matters.

Scenario Rationale

A material scenario should have a concise rationale answering:

Why does this scenario require simulation consideration?

The rationale should connect the scenario to the governed simulation inquiry.

This prevents Scenario Registers from becoming unexplained catalogs.

Scenario Applicability

Not every scenario applies to every: object version, configuration,
environment, population, mission phase, operating mode, or Simulation Scope.

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Canonical source: https://originopenfoundation.org/SCIM.html

SCIM therefore establishes applicability boundaries where necessary.

Scenario Dependency

A scenario may depend upon another condition or event.

For example:

Backup Navigation Failure

may become relevant only after:

Primary Navigation Failure.

The dependency should remain visible where it affects Scenario Design.

Scenario Relationship

Scenarios may relate through: prerequisite, sequence, common cause, escalation, dependency, recovery, variation, alternative state, shared boundary, or shared scenario family.

These relationships can later become important when Scenario Variation and Scenario Coverage are evaluated.

Scenario Family

Related scenarios may be grouped into a Scenario Family without losing individual scenario identity.

For example:

Communication Degradation

may include:

Latency Increase

Intermittent Connection

Bandwidth Reduction

Partial Data Loss

Complete Communication Loss

The family provides structural organization.

Each materially distinct scenario remains independently governable.

Scenario Inclusion

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A candidate scenario should enter the Scenario Register where its material relevance has been sufficiently established.

Inclusion may be:

Required

Conditionally Required

Exploratory

or another governed status appropriate to the simulation.

Scenario Exclusion

A candidate scenario may be excluded where it is: outside Simulation Scope, materially irrelevant, impossible under the governed Simulation Object, equivalent to another governed scenario, duplicative without additional value, or unsupported by the defined simulation inquiry.

Material exclusions should remain documented where later users could otherwise assume the scenario was considered or covered.

No Convenience Exclusion Principle

SIMULOS® establishes:

A materially relevant scenario must not disappear from Scenario Design merely because it is difficult, expensive, computationally demanding, inconvenient, or currently impossible to simulate.

Where the scenario is material but cannot currently be simulated, the correct result may be:

Scenario Required — Simulation Capability Gap

rather than exclusion.

Scenario Duplication

Scenario duplication can create the appearance of broad coverage without expanding the actual condition space examined.

SCIM therefore distinguishes:

distinct scenario

from:

duplicate or materially equivalent scenario.

This becomes important later when Scenario Coverage is assessed.

Scenario Register

SCIM produces a governed Scenario Register containing the identified scenario population relevant to the simulation inquiry.

A Scenario Register may preserve: Scenario Identifier, Scenario Requirement, scenario name, classification, rationale, origin, materiality, applicability, boundary, dependencies, relationships, scenario family, inclusion status, exclusion status, version, and lifecycle status.

The Scenario Register becomes the primary structured input to subsequent SSS modules. Scenario Register ≠ Scenario Set

SCIM distinguishes:

Scenario Register

from:

Governed Simulation Scenario Set.

The Scenario Register identifies and organizes materially relevant scenarios.

The final Scenario Set emerges only after scenarios have been: varied, edge-tested, stress-designed, and coverage-assessed

through subsequent SSS governance spaces.

Scenario Discovery

Scenario identification does not necessarily end before simulation execution.

New scenarios may be discovered through: unexpected simulation behavior, newly identified dependencies, operational incidents, changed environments, new evidence, system modifications, or emerging interactions.

SCIM therefore supports lifecycle scenario discovery.

Emergent Scenario

An Emergent Scenario is a materially relevant scenario identified after the original Scenario Register was established.

It should receive: identity, rationale, origin, classification, materiality assessment, and lifecycle status

like any other governed scenario.

Historical Scenario Preservation

Adding a newly discovered scenario must not create the false impression that earlier simulation campaigns already covered it.

SIMULOS® establishes:

New scenario discovery must preserve the historical boundary of prior Scenario

Coverage.

Scenario Versioning

A scenario may change because of: object change, environment change, parameter change, new knowledge, changed assumptions, dependency change, or refined scenario definition.

Materially changed scenarios should remain version-distinguishable.

Scenario Lifecycle Status

A scenario may move through statuses such as:

Candidate

↓

Identified

↓

Accepted

↓

Configured

↓

Execution Ready

↓

Executed

↓

Reassessment Required

↓

Retired

The exact implementation may differ.

The important governance requirement is that current scenario status remains visible.

Scenario Identification Gap

A Scenario Identification Gap exists where a materially relevant condition appears to exist but cannot yet be sufficiently identified, bounded, or characterized.

A gap should not automatically become an invented scenario.

The correct governance state may remain:

Scenario Identification Incomplete.

Unknown Scenario Space

Complex systems may contain conditions that cannot be fully enumerated in advance.

SCIM does not claim that governance can identify every future scenario.

Instead, it requires that: known scenario requirements are governed, material uncertainty remains visible, discovery pathways exist, and later emerging scenarios can enter the lifecycle without rewriting historical records.

No Exhaustive-Knowledge Claim

SIMULOS® establishes:

A governed Scenario Register demonstrates what has been identified; it does not prove that every possible scenario has been discovered.

This distinction becomes especially important for autonomous, adaptive, multi-agent, and open-environment systems.

Autonomous-System Scenario Identification

Autonomous systems can generate scenario requirements from interactions between: perception, decision state, environment, other agents, human behavior, communication, permissions, autonomy level, resource availability, and system objectives.

A material scenario may therefore arise from a combination rather than one isolated failure.

For example:

Reduced Sensor Confidence + Conflicting External Agent Behavior + Communication Delay + Autonomous Decision Requirement

may represent a distinct scenario requiring governance even if each individual condition has already been identified separately.

Space and Mission Scenario Identification

Space and mission systems make Scenario Identification particularly important because operational testing may be: physically impossible, prohibitively expensive, mission-limited, irreversible, or unable to reproduce rare combinations after deployment.

A spacecraft simulation may therefore need to identify conditions involving: mission-phase transitions, communication loss, navigation uncertainty, thermal extremes, partial power availability, subsystem degradation, timing deviations, ground-segment limitations, and interacting failures.

SCIM does not prescribe these scenarios universally.

It provides the governance structure through which mission-specific conditions can become identifiable and traceable Scenario Requirements.

Scenario Identification for Existing Simulation Programs

SCIM can also be applied retrospectively.

An organization may compare an existing simulation program against the module and ask:

Which Scenario Requirements are explicit?

Which scenarios have unique identities?

Why was each scenario selected?

Which material conditions were excluded?

Which scenarios are duplicates?

Which scenarios were discovered later?

Can every executed scenario be traced back to a governed reason for its existence?

This can expose simulation weaknesses before changing the model itself.

Module Boundary

SCIM begins when the governed Simulation Object, Purpose & Scope, Reality Representation, Assumptions, Model & Environment, and Input & Data Basis provide sufficient context to identify materially relevant simulation conditions.

SCIM ends when candidate conditions have been sufficiently: identified, materiality-assessed, distinguished, bounded, classified, related, included or excluded, and recorded

to establish a governed Scenario Register.

SCIM does not determine the complete Scenario Set.

Subsequent modules govern:

SCVM — Scenario Variation

SECM — Scenario Edge Cases

SSCM — Scenario Stress Conditions

SCCM — Scenario Coverage

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Minimum Implementation Framework

1. Establish Scenario Identification Basis

Identify the governed simulation elements from which Scenario Requirements may arise. 2. Identify Candidate Conditions

Capture materially plausible: states, events, interactions, boundaries, failures, transitions, uncertainties, and alternatives.

3. Assess Materiality

Determine whether each candidate could materially affect the Simulation Purpose, system behavior, simulation outputs, or downstream interpretation. 4. Establish Scenario Identity

Assign sufficient identity, boundary, rationale, origin, classification, applicability, and relationship information to each material scenario. 5. Establish the Scenario Register

Record included, conditionally included, exploratory, excluded, duplicate, unresolved, and newly discovered scenarios while preserving their lifecycle continuity.

Governance Outputs

SCIM may produce: Scenario Identification Basis, Scenario Requirement, Candidate Scenario Record, Scenario Materiality Assessment, Scenario Identifier, Scenario Identity Record, Scenario Boundary Record, Scenario Classification, Scenario Origin Record, Scenario Rationale, Scenario Applicability Record, Scenario Dependency Map, Scenario Relationship Map, Scenario Family, Scenario Inclusion Record, Scenario Exclusion Record, Scenario Equivalence Record, Scenario Duplication Record, Scenario Identification Gap, Scenario Capability Gap Reference, Scenario Discovery Record, Emergent Scenario Record, Scenario Version Record, Scenario Lifecycle Status, and Governed Scenario Register.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

An autonomous-driving simulation program contains thousands of road scenarios.

Most are variations of normal traffic.

SCIM analysis reveals that scenario identification was largely derived from available driving data rather than from a governed Scenario Requirement process. Application

The program identifies additional material conditions involving: degraded perception, conflicting road-user behavior, partial map inconsistency, communication limitations, emergency vehicle interaction, unusual transitions, and combinations of these conditions.

Each receives an explicit Scenario Requirement and Scenario Identity. Result

The primary weakness was not insufficient simulation volume.

It was an incomplete Scenario Register.

The organization can now determine which missing conditions require variation, edge-case design, stress testing, and coverage analysis. Use Case 2 — Spacecraft Mission Simulation

Scenario

A mission simulation extensively covers nominal orbital operations and individually modeled subsystem failures.

SCIM identifies a missing Scenario Requirement:

communication degradation during autonomous recovery while power availability is constrained.

Each condition had previously existed elsewhere in the simulation program, but their operational relationship had never been identified as a distinct scenario. Application

SCIM establishes the combined condition as a governed Simulation Scenario with: Scenario Identity, mission-phase applicability, dependency relationships, rationale, and explicit origin.

Result

A previously invisible mission condition becomes part of the Scenario Register before scenario stress and coverage analysis.

The architecture has not predicted whether the spacecraft will fail.

It has identified a condition that the simulation program should no longer leave structurally invisible. Use Case 3 — Industrial Autonomous Robot

Scenario

An industrial robot has simulation scenarios for: normal production, sensor failure, human proximity, and reduced actuator performance.

An operational incident later reveals that partial sensor degradation combined with unexpected human movement produces behavior not represented in the original scenario population. Application

SCIM creates an Emergent Scenario linked to: the incident, the relevant system configuration, the affected operating state, and the original Scenario Register version.

Result

The new scenario enters future Scenario Design without rewriting historical records.

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Earlier simulations remain correctly represented as:

Scenario Not Previously Identified / Not Previously Covered.

Architectural Position

Scenario Identification Module (SCIM) is the first internal governance space of the Simulation Scenario Standard (SSS).

The complete SSS progression is:

Scenario Identification

↓

Scenario Variation

↓

Scenario Edge Cases

↓

Scenario Stress Conditions

↓

Scenario Coverage

SCIM establishes the governed scenario population upon which all subsequent Scenario Design governance depends.

Without SCIM, scenario variation can vary the wrong scenarios, edge-case analysis can search the wrong boundaries, stress testing can stress irrelevant conditions, and coverage analysis can measure a scenario population whose completeness was never methodologically established.

Foundational Principle

A simulation cannot deliberately examine a materially relevant condition until that condition has first become an identifiable, bounded, and governable Simulation Scenario.

Canonical Closing Statement

Scenario Identification Module establishes the entry point into governed Simulation Scenario Design by transforming materially relevant conditions, states, events, interactions, uncertainties, failures, transitions, dependencies, and alternatives into identifiable and traceable Simulation Scenarios. Through Scenario Requirements, materiality assessment, identity, boundaries, classification, origin, rationale, applicability, dependency mapping, inclusion and exclusion governance, scenario families, discovery, versioning, and Scenario Register formation, SCIM ensures that simulation programs do not confuse large numbers of available scenarios with systematic identification of the conditions that actually matter to the simulation inquiry.

Modul 2

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